THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A circular coil carrying current has a south pole facing the viewer. The current as seen is 1

(a) clockwise **(b)** alternating **(c)** zero **(d)** anticlockwise

2. The number of reversals of AC in a minute of 50 Hz supply is 1

(a) 100 **(b)** 3000 **(c)** 600 **(d)** 6000

3. Assertion (A): A conductor moving parallel to the magnetic field induces no current. Reason (R): There is no change in the magnetic field lines linked with the conductor. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. The right-hand thumb rule applied to a vertical wire carrying current upward gives the field direction 1

(a) downward **(b)** anticlockwise when seen from above
(c) upward **(d)** clockwise when seen from above

5. A fuse of 5 A is connected in a circuit with a 1.5 kW heater at 220 V. The fuse 1

(a) gives no protection **(b)** does not blow
(c) blows because I = 1.5 A **(d)** blows because I = 6.8 A

6. Assertion (A): AC is preferred for the transmission of power over long distances. Reason (R): Much less energy is lost when AC is sent over long distances. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A positively charged particle moves along the direction of the magnetic field. The force on it is 1

(a) zero **(b)** maximum
(c) along the field **(d)** perpendicular to the field

8. Appliances A (1 kW), B (0.5 kW) and C (0.05 kW) use the same 220 V line protected by a 5 A fuse. Which combination will blow the fuse? 1

(a) A and C together **(b)** B and C together **(c)** A alone **(d)** A and B together

9. Assertion (A): The earth wire is connected to the metal body of a geyser. Reason (R): It safeguards the user by taking leakage current to the earth. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. A uniform magnetic field exists in a region. A charged particle enters along the field lines. The particle 1

(a) moves in a parabola **(b)** moves in a straight line

(c) stops **(d)** moves in a circle

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What happens to the induced current if the same magnet is pushed into a coil with a larger number of turns? 2

12. What is the function of the split-ring commutator in an electric motor? 2

13. How would you test whether a wire in a circuit is carrying current without touching it, using a compass? 2

14. Why does a magnet attract an iron nail but not a copper wire? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A heater of 2 kW and a toaster of 1 kW are used at a time on a 5 A fuse circuit at 220 V. Will the fuse blow? Show by calculation. 3

16. How can you determine the polarity of an electromagnet using a compass needle? 3

17. Explain why an alternating current is produced by the coil of a generator when it rotates continuously in the same direction. 3

OR

Explain how a magnet, a coil and a galvanometer can be used to show that the induced current depends on the speed of the magnet. 3

18. Under what circumstances is the force on a moving charge in a magnetic field maximum? 3

OR

How does the output of an AC generator change if the number of turns in the coil is doubled? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain with the principle and steps how a magnetic field is produced by a current in a solenoid and how it can be made 10 times stronger. 5

OR

State the frequency, the period and the typical voltage of the household supply in India. A device draws 5 A at 220 V; find its power and the energy used in 3 hours in kWh. 5

20. A student has a solenoid, a bar magnet, a galvanometer and a battery. Design activities to show 5

(a) the magnetic effect of current

(b) electromagnetic induction. State the observations and the conclusions.

OR

Describe the effect on the heating and safety of an appliance if 5

(a) the earth wire is not connected

(b) the fuse is placed in the neutral wire

(c) the switch is placed in the neutral wire.

END OF QUESTION PAPER Total: 40 Marks